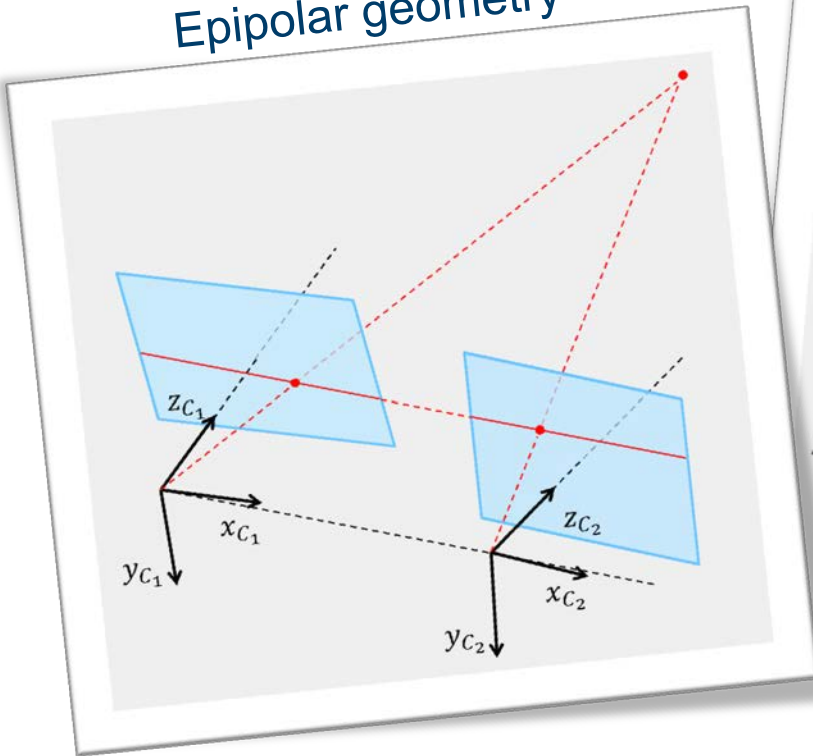


Lecture 6.1

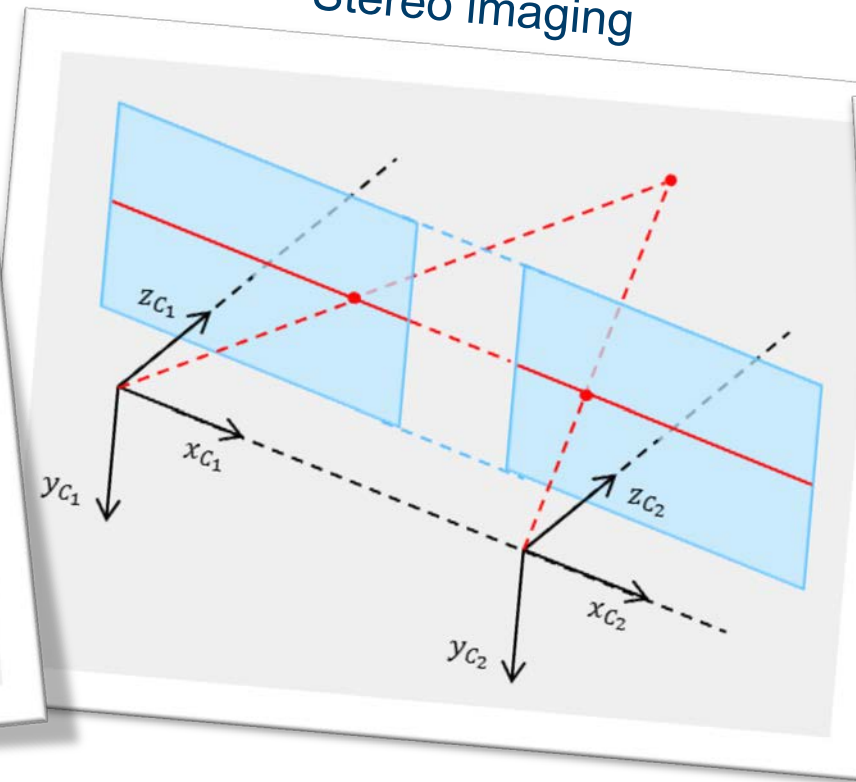
Basic epipolar geometry

Weekly overview – Stereo imaging

Epipolar geometry



Stereo imaging

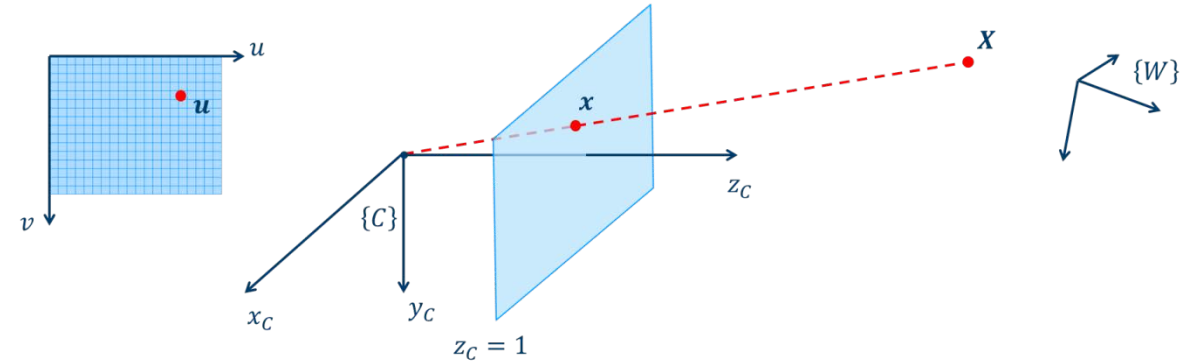


Stereo processing



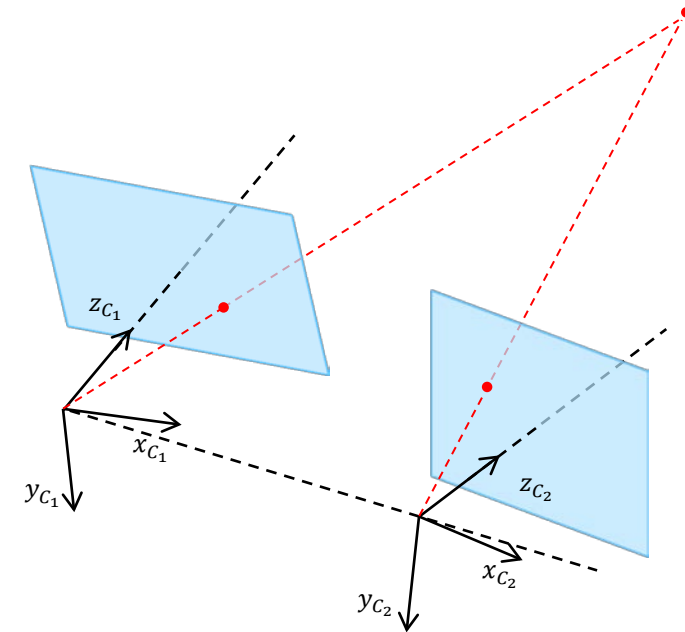
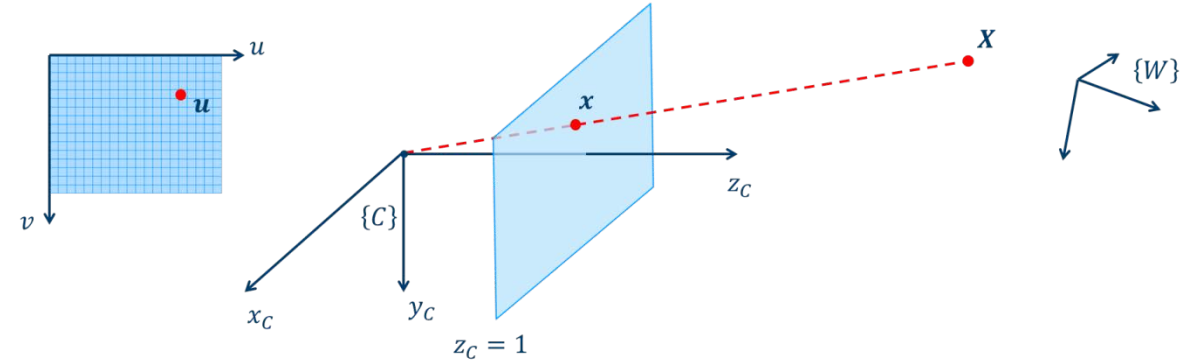
Introduction

- Single-view geometry
 - Camera model $P\tilde{X} = \tilde{u}$
 - Finite projective camera $P = K[R \quad t]$
 - Undistortion
 - Estimating P from 3D-2D correspondences
 - Calibration
 - PnP



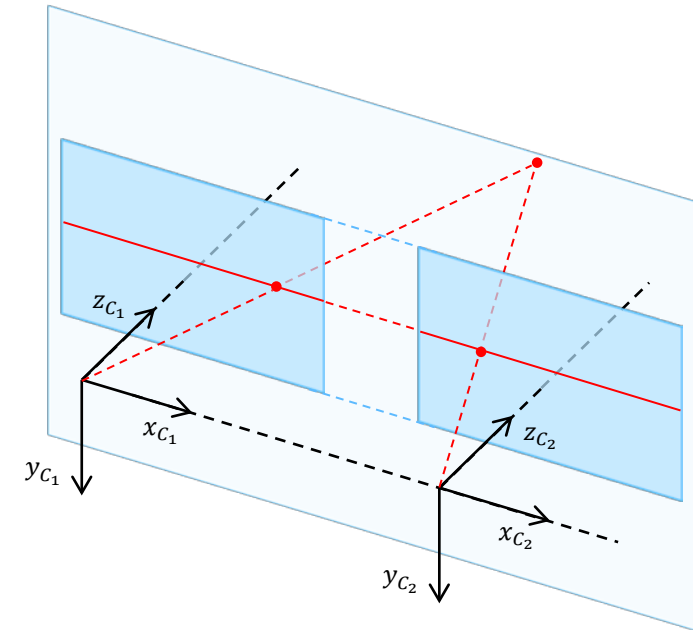
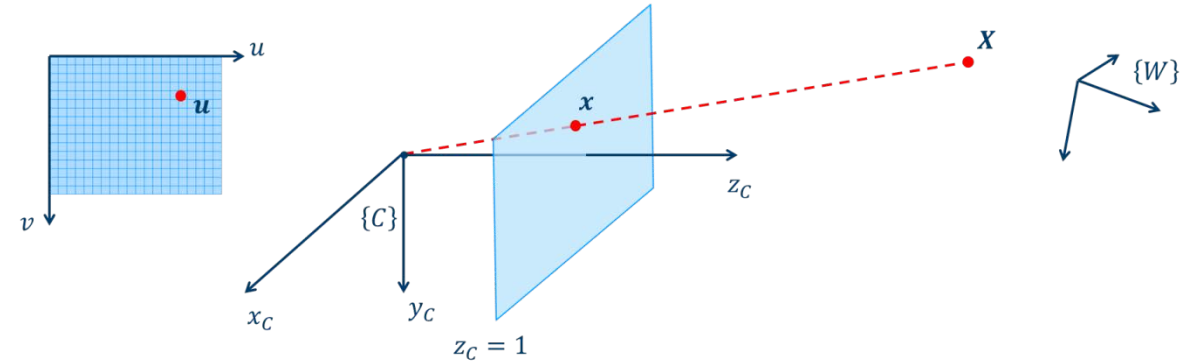
Introduction

- Single-view geometry
 - Camera model $P\tilde{X} = \tilde{u}$
 - Finite projective camera $P = K[R \ t]$
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 - Calibration
 - PnP
- Two-view geometry
 - Epipolar geometry is the geometric relationship between two perspective cameras
 - Two camera models $P_1\tilde{X} = \tilde{u}_1$, $P_2\tilde{X} = \tilde{u}_2$
 - Next week - General two-view



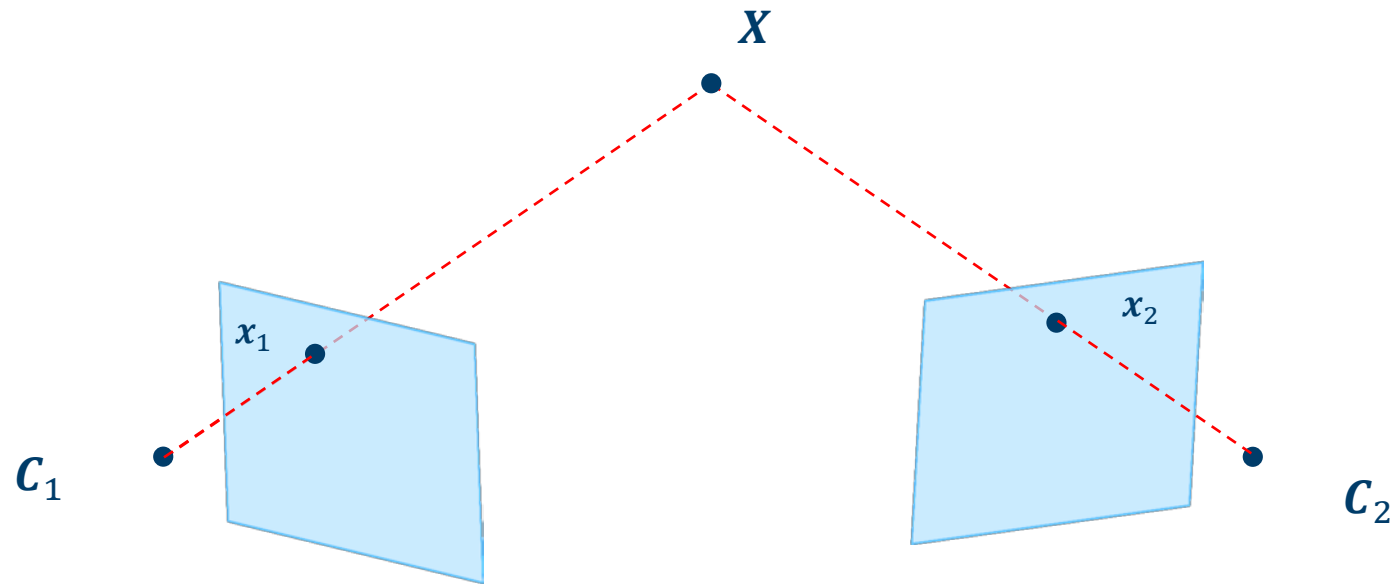
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 - Next week - General two-view
 - This week - Stereo view



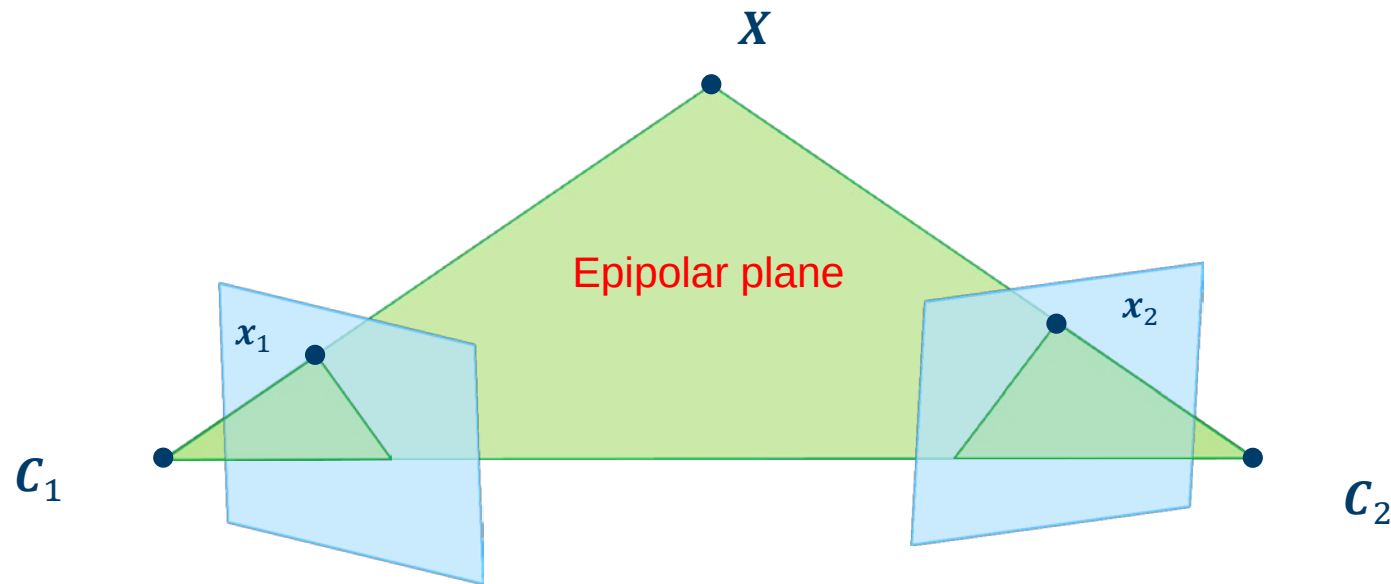
Epipolar geometry

- Two-view geometry involve several new geometrical entities compared to single-view geometry



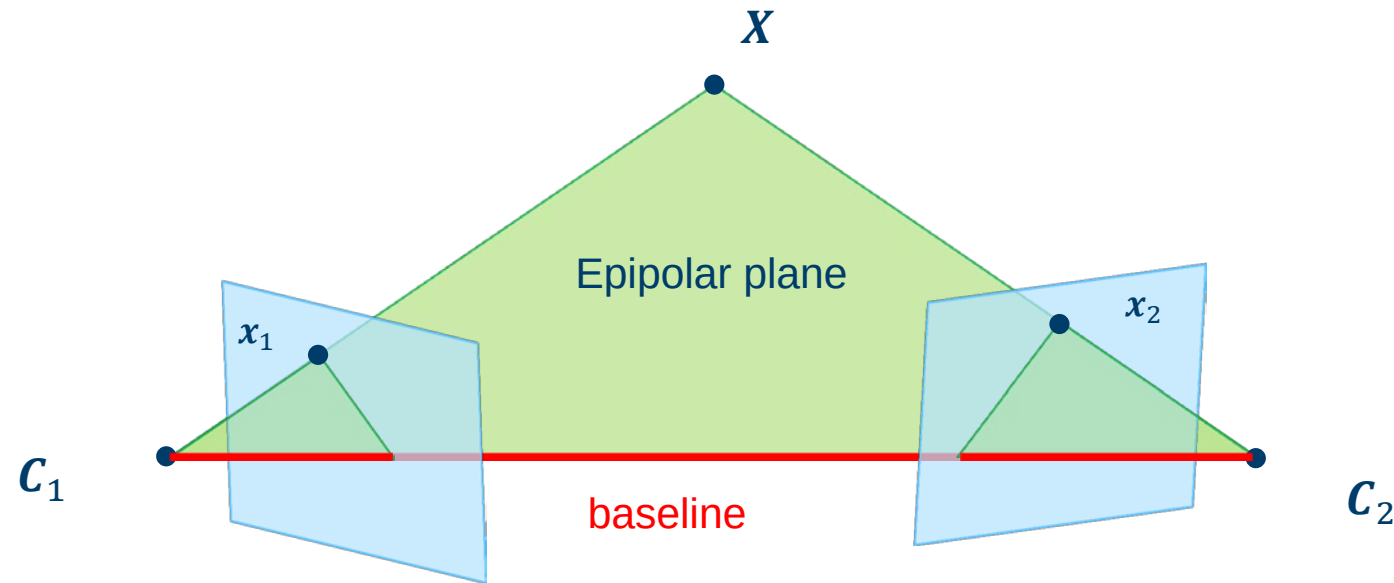
Epipolar geometry

- Two-view geometry involve several new geometrical entities compared to single-view geometry
- The **epipolar plane** is the plane containing X and the two camera centers C_1 and C_2



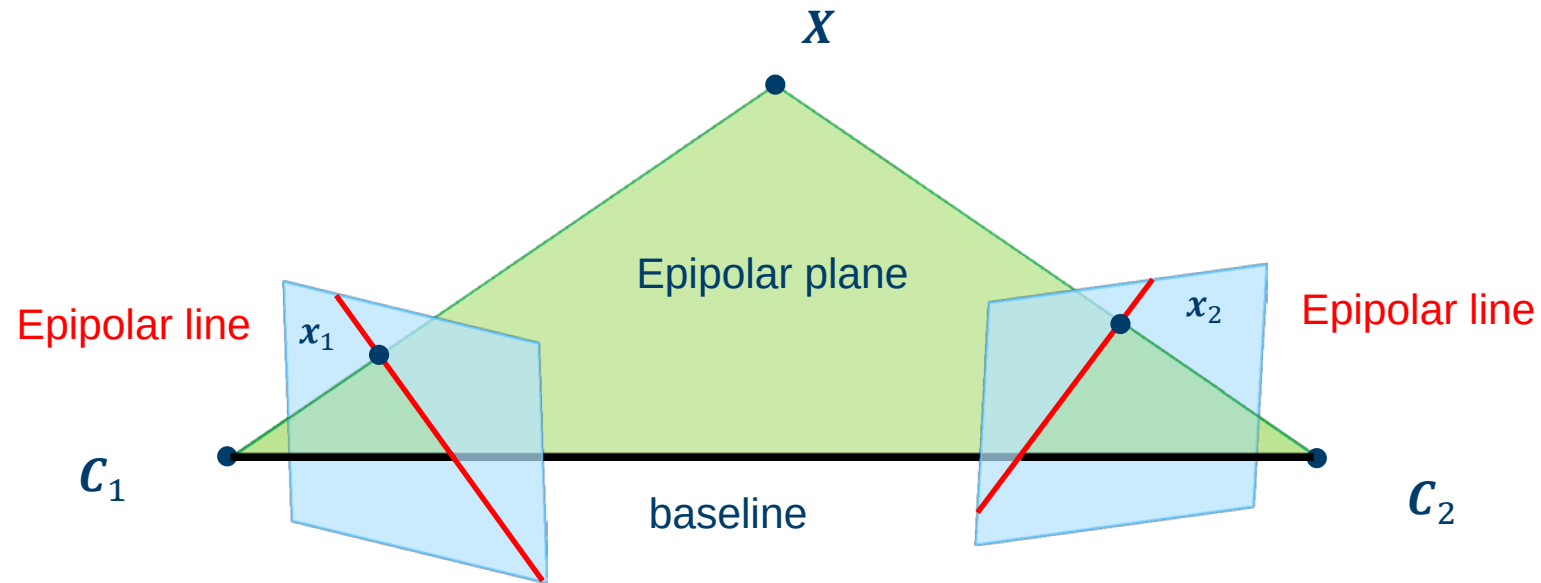
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- Two-view geometry involve several new geometrical entities compared to single-view geometry
- The **epipolar plane** is the plane containing X and the two camera centers C_1 and C_2
- The **baseline** is the line joining the two camera centers



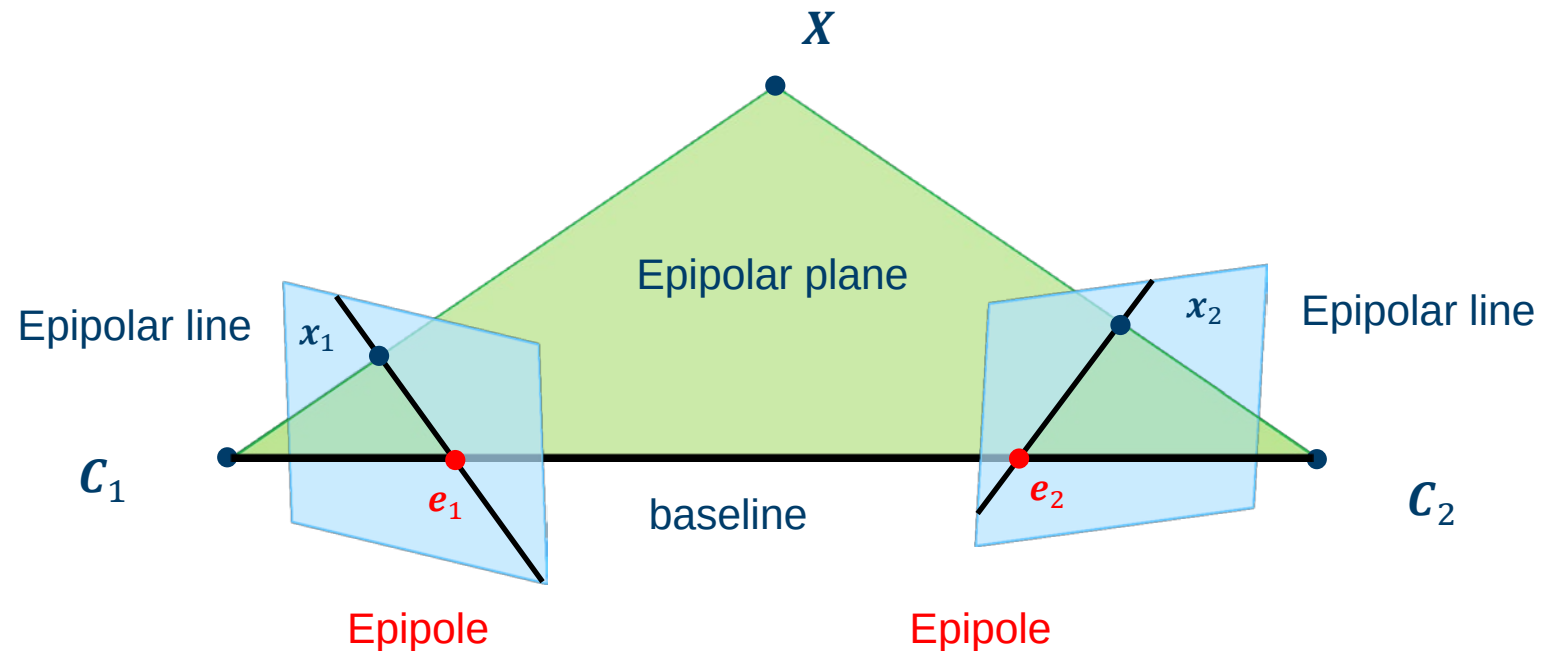
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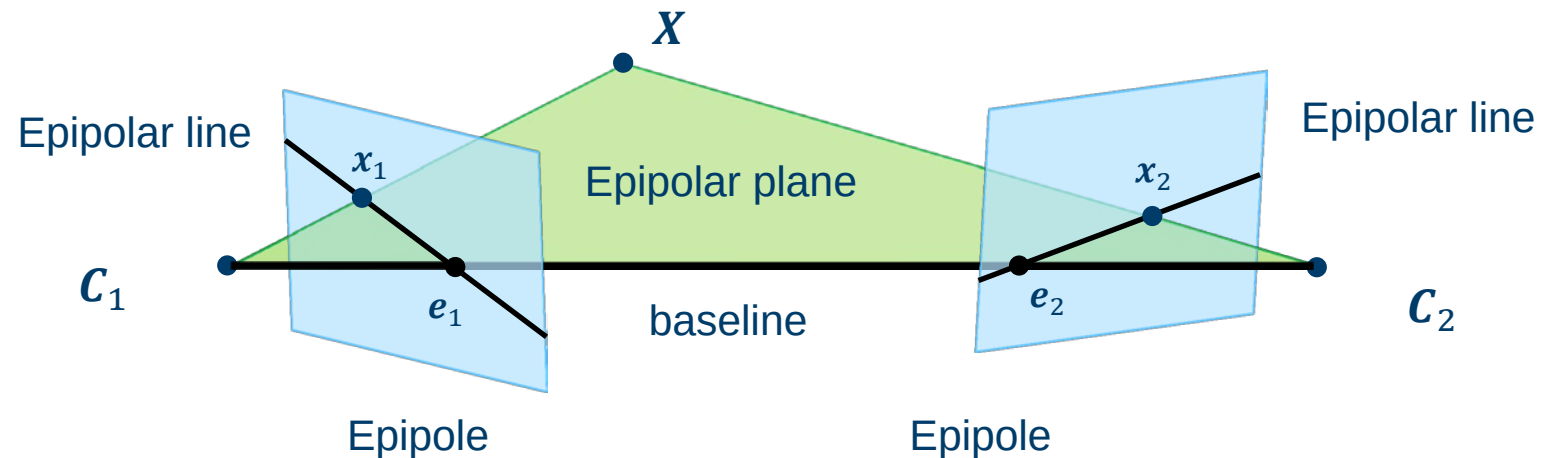
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Epipolar geometry

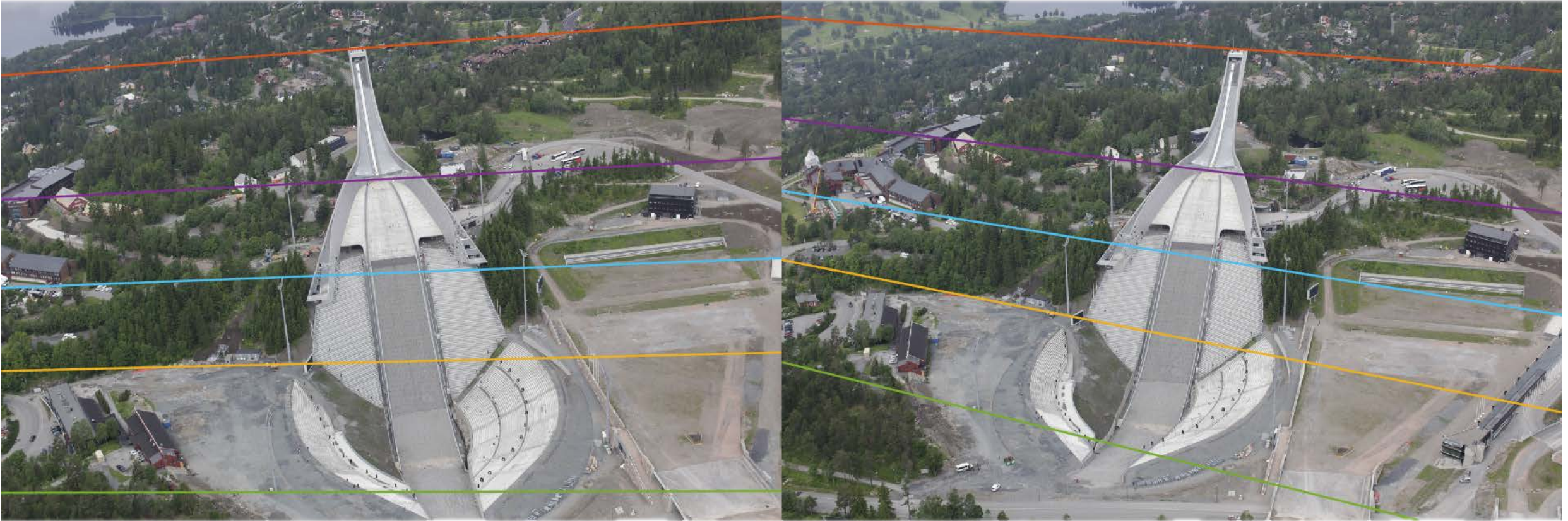
- Two-view geometry involve several new geometrical entities compared to single-view geometry
- The **epipolar plane** is the plane containing X and the two camera centers C_1 and C_2
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- The **epipolar lines** are where the epipolar plane intersect the image planes
- The **epipoles** are where the baseline intersects the two image planes
- The baseline and epipoles are uniquely defined by the two camera matrices P_1 and P_2
- The epipolar plane and epipolar lines depends on the observed point X



Example



Example

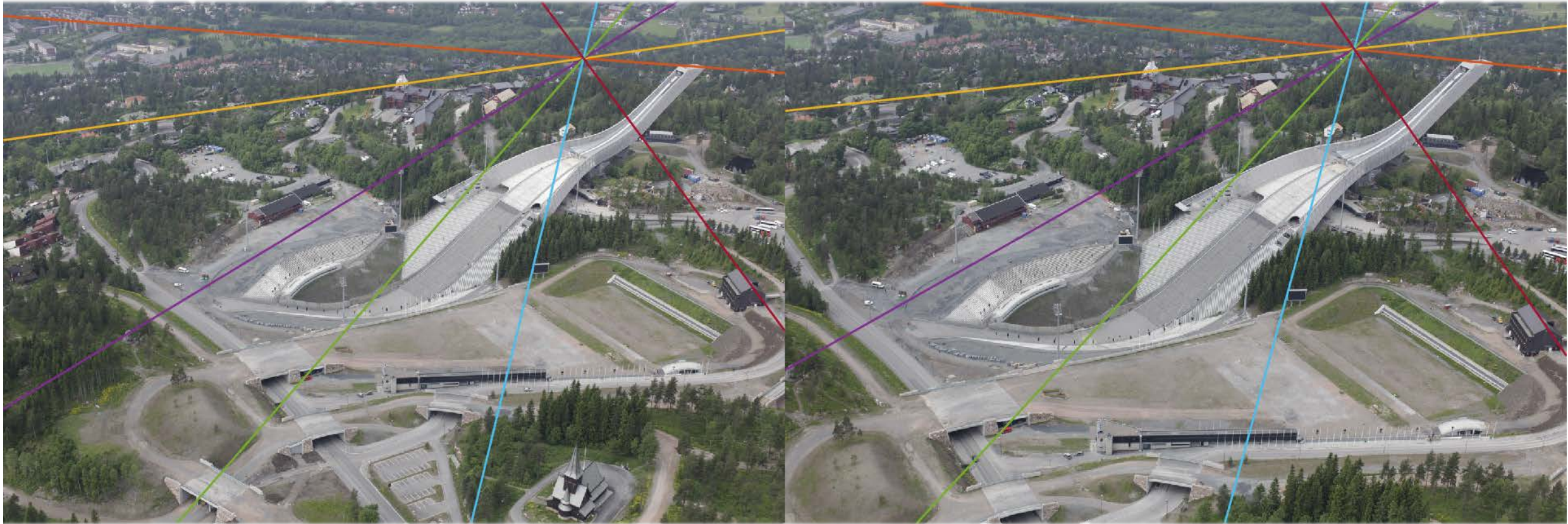


- Corresponding points lie on corresponding epipolar lines
- Both epipoles are outside of the visible part of the image planes

Example



Example



- Corresponding points lie on corresponding epipolar lines
- Both epipoles are visible as the intersection of epipolar lines

Summary

- Epipolar geometry
 - Epipolar planes
 - Epipolar lines
 - Epipoles
- Topics ahead
 - Stereo imaging
 - Representing epipolar geometry
 - Estimating epipolar geometry
 - 3D from epipolar geometry
 - Relative pose from epipolar geometry
 - More views...

- Additional reading:
 - Szeliski: 11 introduction & 11.1

